

Identifying Factors of Airbnb Rental Prices in San Diego

Prabhleen Gill

SOECS

University of the Pacific
Stockton, United States of America
p_gill9@u.pacific.edu

Sophie Tran

SOECS

University of the Pacific
Stockton, United States of America
s_tran39@u.pacific.edu

Viet Anh Tu

SOECS

University of the Pacific
Stockton, United States of America
v_tu2@u.pacific.edu

Abstract—This project analyzes Airbnb rental listings in San Diego to identify key factors influencing pricing, and to develop an interactive tool for exploring these patterns. Using data from Inside Airbnb, the analysis focuses on listing characteristics such as room type, property type, number of bedrooms and bathrooms, and location. After preprocessing the dataset, exploratory analysis was performed to examine the relationships between these variables and price. The results show that larger listings and entire homes tend to have higher prices, while location significantly affects both price levels and variability across neighborhoods. These findings were incorporated into an interactive dashboard built in Tableau, which allows users to filter listings and explore price patterns geographically. By combining data analysis with user-focused visualization, this project provides insights into Airbnb pricing dynamics and offers a practical tool to support informed decision-making for potential users.

Keywords—*Airbnb, rental price determinants*

I. INTRODUCTION

The rapid growth of the sharing economy has significantly transformed the hospitality industry, with Airbnb emerging as one of the most dominant platforms for short-term accommodation rentals. Pricing on Airbnb is highly dynamic and influenced by a wide range of factors, including property characteristics, location, room type, amenities, and host-related attributes. Understanding the determinants of Airbnb rental prices is important not only for hosts seeking to optimize their pricing strategies but also for travelers, policymakers, and urban planners interested in market transparency and housing dynamics.

This project focuses on identifying and analyzing the key factors that influence Airbnb rental prices in San Diego.

II. RELATED WORK

Researchers have studied Airbnb pricing determinants in past works using different methods. While we studied and found determinants of Airbnb prices through graphs and data analysis, other papers have used a multivariable analysis technique that can predict Airbnb listing prices [8] or spatial regression models [2]. OLS (ordinary least squares) and GWR (geographically weighted regression) have been used to

analyze Airbnb price factors such as season, location, and property/room type [3]. Another paper has built on the spatial regression technique by developing a new technique called Multiscale Geographically Weighted Regression (MGWR) [1]. Our project builds upon these past research papers by analyzing Airbnb price listings through different features, such as room type, property type, and location (neighborhoods), and determining which features factor into Airbnb prices and how. Our project is slightly different from prior works, since we present our results on an interactive dashboard for users, which was created by using a data visualization tool called Tableau.

III. IMPLEMENTATION DETAILS

The dataset used in this project was obtained from the Inside Airbnb website, focusing on listings specifically from San Diego. It included information such as listing name and description, neighborhood, host information, property type, room type, number of accommodations, number of bathrooms, number of bedrooms, and beds, available amenities, price, minimum and maximum nights, number of reviews, and rating scores. This information was used to explore factors influencing Airbnb prices and to build an interactive dashboard for analysis.

The analysis was performed within a Google Colab notebook using Python as the primary programming language. Pandas and Numpy were used primarily for preprocessing and modifying the dataset, while Matplotlib and Seaborn were used to produce visualizations within the notebook. Before analysis, the dataset went through several preprocessing steps to improve usability and ensure consistency. The price variable was cleaned by removing dollar signs and converting it to a numeric format. Rows with missing data values were removed to maintain data quality. Categorical variables were standardized for clarity. For example, room types were simplified into four categories: entire, private, shared, and hotel. Property types were cleaned, normalized to lowercase, and grouped into broader categories such as house, apartment, guest property, hotel, and unique stays. Rare property types with low frequency were grouped into an “other” category.

Extreme outliers in price were removed to prevent distortion in the results.

Additional feature engineering was performed to enhance the dataset for deeper analysis. The accommodates variable, which represents the number of guests a listing can host, was grouped into ranges to simplify analysis and visualization. The amenities field, originally stored as a list of strings, was converted into a Python list and then transformed into a numeric feature representing the total number of amenities for each listing. Binary variables were converted into numeric variables. Other numeric variables, including number of reviews, minimum nights, and availability, were also converted to consistent numeric formats.

To better interpret price variation, K-means clustering was applied to group listings into six price tiers. These tiers were Low, Lower-Mid, Mid, Upper-Mid, High, and Very High. This helped to improve the visualization of prices in the dashboard as well as display the distribution of prices across the listings in the dataset. Since the cluster labels produced by K-means were arbitrary, the clusters were reordered by average price to create meaningful categories ranging from low to very high. These categories were then used in the dashboard to visually distinguish listings by relative cost.

An interactive dashboard was developed using Tableau to present the findings. The dashboard was designed with the purpose of usability, placing a map of the San Diego listings at the center and surrounding it with supporting visualizations. These included average price by room type, average price by property type, and price ranges of neighborhoods. Users can filter the data on the map by room type, property type, neighborhood, and key numeric attributes such as price, number of bedrooms, bathrooms, and beds, and rating. The charts presented in the dashboard allow the user to explore how various factors affect the prices of Airbnb listings and make comparisons based on their preferences.

The proposed solution is designed to remain effective as the size of the dataset increases. The preprocessing pipeline relies on efficient, vectorized operation in Python using the pandas library, which can handle large datasets with minimal modifications. Feature engineering steps can scale linearly with the number records, ensuring that the processing time increases proportionally. The clustering approach operates on a single numerical variable, making it computationally efficient for larger datasets. Additionally, the visualization component is implemented in Tableau, which is capable of handling large datasets through extracts and aggregation techniques. By relying on aggregated data rather than raw data, the dashboard can remain responsive and interpretable. Overall, the combination of techniques used in producing these results ensures that the solution can handle increased data volume without significant performance degradation.

A. Results

The exploratory analysis was performed through visualizations of the data provided in the listings dataset. The price of listings was compared with several factors, including room type, property type, number of bathrooms, number of bedrooms, and neighborhoods. Figure 1 shows the average

price by room type. Entire rooms have the highest average price of all categories, indicating that privacy and exclusivity are major drivers of cost.

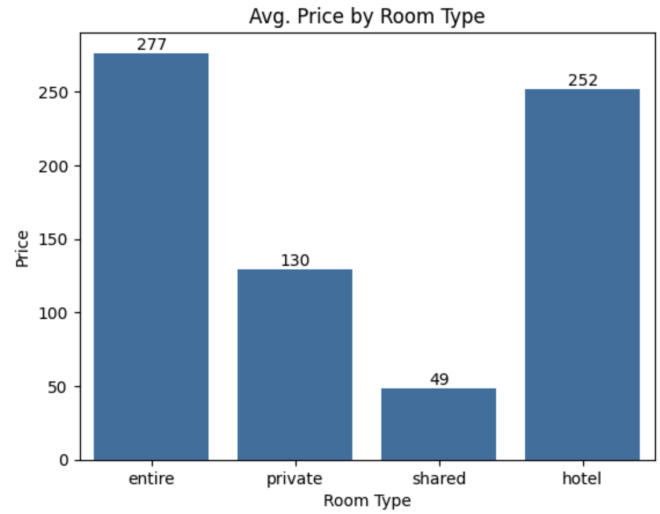


Fig. 1. Average price by room type.

Figure 2 shows price by number of bedrooms. There is a clear upward trend as price increases with the number of bedrooms. The chart also includes a confidence interval band that is centered mostly around 8 to 10 bedrooms. This indicates that there is a greater variability of prices as the number of bedrooms increases as well as fewer observations for that range.

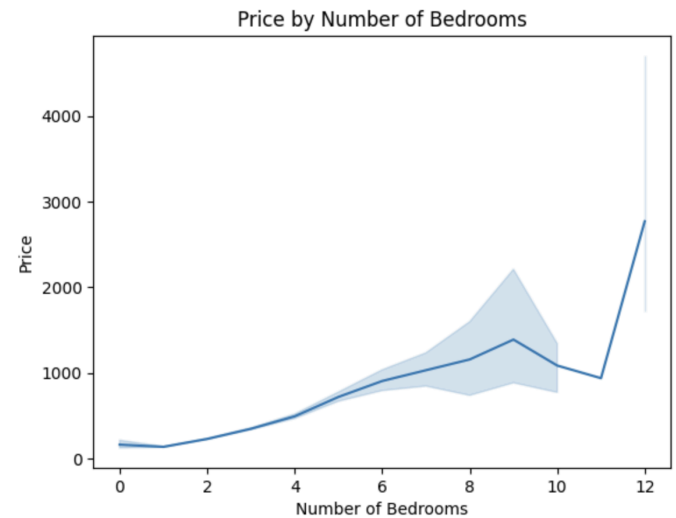


Fig. 2. Price by number of bedrooms.

Figure 3 shows the boxplots of prices for the top 10 most expensive neighborhoods based on average price. Differences in median price levels as well as overall distribution of prices are evident. La Jolla has the largest price range, with prices reaching up to around \$1200. It was followed by Mission Bay, which had prices reaching up to around \$900. The remaining eight neighborhoods did not have prices beyond \$600, but were still notably more expensive compared to other neighborhoods.

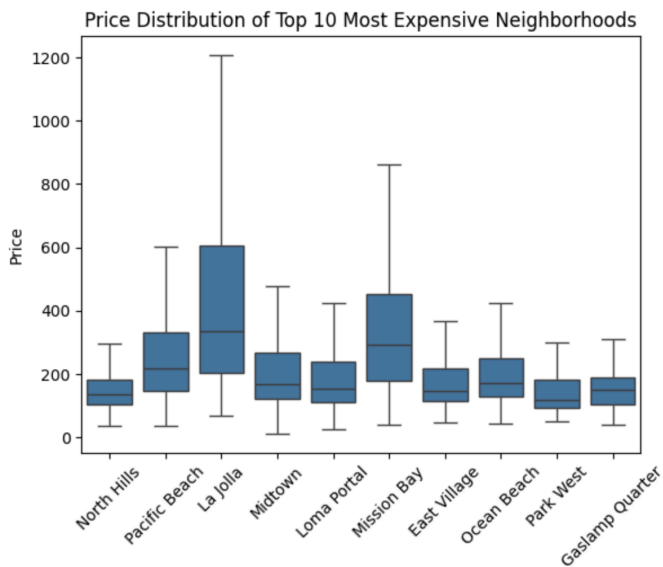


Fig. 3. Price distribution of top 10 most expensive neighborhoods

The final dashboard (Figure 4) included visualizations for average price by room type, average price by property type, price ranges for neighborhoods, and a map displaying the listings as well as relevant information for each listing. These visualizations were chosen because they are user-friendly and do not require prior knowledge of statistics to understand, making it more accessible for a broader range of users. Users can use the filters to adjust the map to their preferences, allowing a more personalized exploration of price factors. The interactive dashboard can be accessed through this link: https://public.tableau.com/shared/CYXWJ4ZO8?display_count=n&origin=viz_share_link

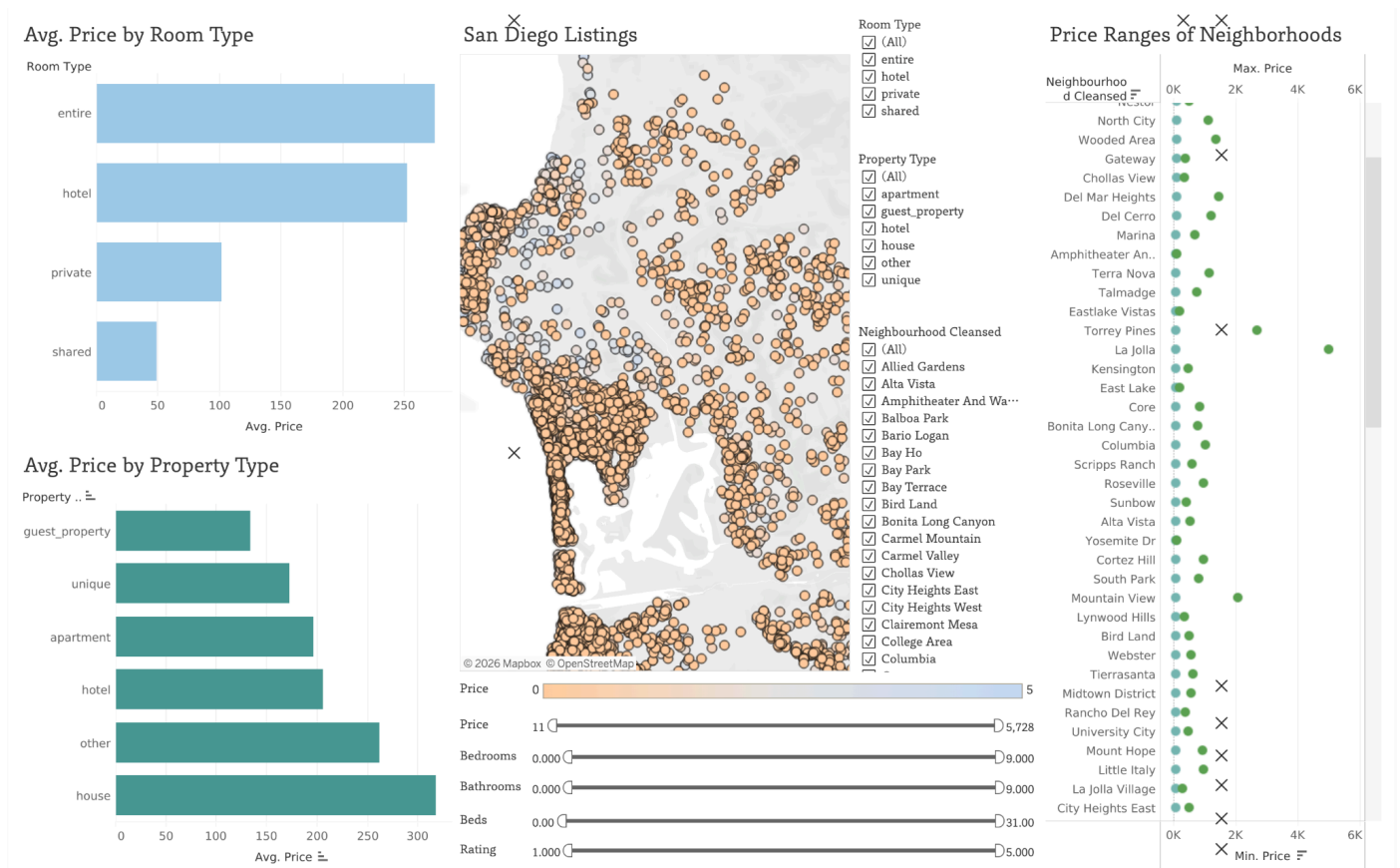


Fig. 4. Interactive dashboard for exploring price factors for Airbnb listings in San Diego

IV. CONCLUSION

The interactive Tableau dashboard created in this project allows users to easily explore Airbnb pricing patterns by applying filters based on their preferences. This dashboard is useful for hosts who want to set competitive prices, travelers who are searching for budget accommodations, and researchers who are interested in short-term rental markets.

Overall, this project adds to existing Airbnb pricing research by focusing on visual and exploratory analysis rather than only using predictive or regression models. Future work could improve this study by adding time-related factors such as seasonality, changes in demand, or host activity over time. Predictive models could also be applied to help estimate optimal pricing strategies.

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